

AIAA2711 (L01) - Mathematics for AI

Final Examination (Answer)

Spring 2026

The Hong Kong University of Science and Technology (Guangzhou)

Type	Part I	Part II	Part III						Total
			1	2	3	4	5	6	
Score									

Part I: True (T) or False (F) Questions.

1	2	3	4	5	6	7	8	9	10
F	T	F	F	T	F	T	T	F	F

Part II: Short answer questions.

1. $\|A - \hat{A}\|_2 = 0.1$

2. $\frac{\partial f}{\partial \mathbf{x}} = 2\mathbf{x}^\top \mathbf{A} = \begin{bmatrix} 2x_1 + 4x_2 & 4x_1 + 2x_2 & 6x_3 \end{bmatrix}$

3. $w_1 = 1.3, b_1 = 0.8$

Explanation: The mean squared error (MSE) loss function for a dataset with N training examples is defined as:

$$L(w, b) = \frac{1}{N} \sum_{i=1}^N (y_i - (wx_i + b))^2$$

For our problem, $N = 2$. We first compute the gradients of the loss with respect to w and b :

$$\begin{aligned} \frac{\partial L}{\partial w} &= \frac{1}{N} \sum_{i=1}^N 2(y_i - (wx_i + b))(-x_i) = -\frac{2}{N} \sum_{i=1}^N x_i (y_i - (wx_i + b)) \\ \frac{\partial L}{\partial b} &= \frac{1}{N} \sum_{i=1}^N 2(y_i - (wx_i + b))(-1) = -\frac{2}{N} \sum_{i=1}^N (y_i - (wx_i + b)) \end{aligned}$$

Substitute the initial parameters $w_0 = 0, b_0 = 0$ and the training data:

$$\begin{aligned} \left. \frac{\partial L}{\partial w} \right|_{w=0, b=0} &= -\frac{2}{2} [1 \times (3 - 0) + 2 \times (5 - 0)] = -(3 + 10) = -13 \\ \left. \frac{\partial L}{\partial b} \right|_{w=0, b=0} &= -\frac{2}{2} [(3 - 0) + (5 - 0)] = -(3 + 5) = -8 \end{aligned}$$

Apply the batch gradient descent update rule $\theta_{\text{new}} = \theta_{\text{old}} - \eta \cdot \nabla L(\theta_{\text{old}})$:

$$\begin{aligned}w_1 &= w_0 - \eta \cdot \frac{\partial L}{\partial w} = 0 - 0.1 \times (-13) = 1.3 \\b_1 &= b_0 - \eta \cdot \frac{\partial L}{\partial b} = 0 - 0.1 \times (-8) = 0.8\end{aligned}$$

4. $I(X; Y) \approx 0.034$ nats

5. ≈ 0.322 bits

Explanation: The Kullback-Leibler divergence from distribution P to Q for discrete random variables is defined as:

$$KL(P, Q) = \sum_{x \in \mathcal{X}} P(x) \log_2 \left(\frac{P(x)}{Q(x)} \right)$$

where $\mathcal{X} = \{\text{cat}, \text{dog}\}$ is the sample space. Substitute the given probabilities into the formula:

$$\begin{aligned}KL(P, Q) &= P(\text{cat}) \log_2 \left(\frac{P(\text{cat})}{Q(\text{cat})} \right) + P(\text{dog}) \log_2 \left(\frac{P(\text{dog})}{Q(\text{dog})} \right) \\&= 0.5 \times \log_2 \left(\frac{0.5}{0.8} \right) + 0.5 \times \log_2 \left(\frac{0.5}{0.2} \right) \\&= 0.5 \times \log_2(0.625) + 0.5 \times \log_2(2.5)\end{aligned}$$

Calculate the logarithmic terms:

$$\begin{aligned}-\log_2(0.625) &= \log_2 \left(\frac{5}{8} \right) = \log_2(5) - \log_2(8) \approx 2.3219 - 3 = -0.6781 \\-\log_2(2.5) &= \log_2 \left(\frac{5}{2} \right) = \log_2(5) - \log_2(2) \approx 2.3219 - 1 = 1.3219\end{aligned}$$

Substitute back:

$$\begin{aligned} KL(P \parallel Q) &\approx 0.5 \times (-0.6781) + 0.5 \times 1.3219 \\ &= -0.33905 + 0.66095 \\ &= 0.3219 \\ &\approx 0.322 \text{ bits} \end{aligned}$$

Part III: Long Answer Questions.

Question 1

(a) Let $B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$ be the matrix whose columns are the basis vectors of U . The projection of $\mathbf{v}$ onto L is given by (2 points)

$$\pi_L(\mathbf{v}) = \mathbf{x}_0 + \pi_U(\mathbf{v} - \mathbf{x}_0),$$

where π_U is the orthogonal projection onto U . Compute $\mathbf{v} - \mathbf{x}_0 = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}$. The projection onto U is $\pi_U(\mathbf{z}) = B(B^\top B)^{-1}B^\top \mathbf{z}$.

$$B^\top B = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad (B^\top B)^{-1} = \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}, \quad B^\top(\mathbf{v} - \mathbf{x}_0) = \begin{bmatrix} 5 \\ 6 \end{bmatrix}.$$

(2 points for above) The coordinates are $\boldsymbol{\lambda} = (B^\top B)^{-1}B^\top(\mathbf{v} - \mathbf{x}_0) = \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} \frac{4}{3} \\ \frac{7}{3} \end{bmatrix}$. Hence,

$$\pi_U(\mathbf{v} - \mathbf{x}_0) = B\boldsymbol{\lambda} = \begin{bmatrix} \frac{4}{3} \\ \frac{7}{3} \\ \frac{11}{3} \end{bmatrix}.$$

(2 points for above) Finally,

$$\pi_L(\mathbf{v}) = \mathbf{x}_0 + \pi_U(\mathbf{v} - \mathbf{x}_0) = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} \frac{4}{3} \\ \frac{7}{3} \\ \frac{11}{3} \end{bmatrix} = \begin{bmatrix} \frac{7}{3} \\ \frac{10}{3} \\ \frac{11}{3} \end{bmatrix}.$$

(1 point for above)

(b) (2 point for first equation 1 point for second) The distance is $\|\mathbf{v} - \pi_L(\mathbf{v})\|_2$.

$$\mathbf{v} - \pi_L(\mathbf{v}) = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} - \begin{bmatrix} \frac{7}{3} \\ \frac{10}{3} \\ \frac{11}{3} \end{bmatrix} = \begin{bmatrix} -\frac{1}{3} \\ -\frac{1}{3} \\ \frac{1}{3} \end{bmatrix},$$

$$\|\mathbf{v} - \pi_L(\mathbf{v})\|_2 = \sqrt{\frac{1}{9} + \frac{1}{9} + \frac{1}{9}} = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}.$$

Question 2

- (a) 6 points total:

– 1 point: Let x^* be a local minimum of f and we assume for contradiction that x^* is not a global minimum. Then there exists some $y \in \mathbb{R}^n$ such that $f(y) < f(x^*)$.

– 2 points: Consider the point $z_t = (1 - t)x^* + ty$ for $t \in (0, 1)$. By convexity of f :

$$f(z_t) \leq (1 - t)f(x^*) + tf(y) < (1 - t)f(x^*) + tf(x^*) = f(x^*)$$

– 2 points: For sufficiently small $t > 0$, $\|z_t - x^*\| = t\|y - x^*\| < \epsilon$, which means z_t is in the local neighborhood of x^* .

– 1 point: But we just showed $f(z_t) < f(x^*)$, which contradicts the definition of x^* being a local minimum. Therefore, our assumption is false, and x^* must be a global minimum.

- (b) 4 points total:

– 1 points: Compute the gradient: $\nabla f(w) = \begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix} w + \begin{pmatrix} 2 \\ 1 \end{pmatrix}$

– 1 points: First iteration: $w_1 = w_0 - \eta \nabla f(w_0) = \begin{pmatrix} 0 \\ 0 \end{pmatrix} - 0.2 \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} -0.4 \\ -0.2 \end{pmatrix}$

– 2 points: Second iteration: $\nabla f(w_1) = \begin{pmatrix} 4(-0.4) + 2 \\ 1(-0.2) + 1 \end{pmatrix} = \begin{pmatrix} 0.4 \\ 0.8 \end{pmatrix}$, so $w_2 =$

$$\begin{pmatrix} -0.4 \\ -0.2 \end{pmatrix} - 0.2 \begin{pmatrix} 0.4 \\ 0.8 \end{pmatrix} = \begin{pmatrix} -0.48 \\ -0.36 \end{pmatrix}$$

Question 3

- (a) 2 points total:
 - 1 point: $P(X = -1) = 0.4, P(X = 1) = 0.6$
 - 1 points: $P(Y = -1) = 0.3, P(Y = 0) = 0.3, P(Y = 1) = 0.4$
- (b) 2 points total:
 - 1 point: Recall conditional probability formula: $P(Y = y|X = x) = \frac{P(X=x,Y=y)}{P(X=x)}$
 - 1 points: $P(Y = 1|X = 1) = \frac{0.3}{0.6} = 0.5$
- (c) 4 points total:
 - 1 point: $E[X] = (-1) \times 0.4 + 1 \times 0.6 = 0.2$
 - 1 point: $E[Y] = (-1) \times 0.3 + 0 \times 0.3 + 1 \times 0.4 = 0.1$
 - 2 point: $E[XY] = (-1) \times (-1) \times 0.1 + (-1) \times 0 \times 0.2 + (-1) \times 1 \times 0.1 + 1 \times (-1) \times 0.2 + 1 \times 0 \times 0.1 + 1 \times 1 \times 0.3 = 0.1$
- (d) 2 points total:
 - 1 point: No, X and Y are not independent
 - 1 point: Justification: $P(X = 1, Y = 1) = 0.3 \neq P(X = 1)P(Y = 1) = 0.6 \times 0.4 = 0.24$

Question 4

Solution: We verify that $\langle \cdot, \cdot \rangle_A$ satisfies the axioms of an inner product.

1. **Symmetry:** (3 points) Since A is symmetric ($A^\top = A$), for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^3$,

$$\langle \mathbf{y}, \mathbf{x} \rangle_A = \mathbf{y}^\top A \mathbf{x} = (\mathbf{y}^\top A \mathbf{x})^\top = \mathbf{x}^\top A^\top \mathbf{y} = \mathbf{x}^\top A \mathbf{y} = \langle \mathbf{x}, \mathbf{y} \rangle_A.$$

2. **Bilinearity:** (3 points) For any $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^3$ and $\alpha \in \mathbb{R}$,

$$\langle \alpha \mathbf{x} + \mathbf{y}, \mathbf{z} \rangle_A = (\alpha \mathbf{x} + \mathbf{y})^\top A \mathbf{z} = \alpha \mathbf{x}^\top A \mathbf{z} + \mathbf{y}^\top A \mathbf{z} = \alpha \langle \mathbf{x}, \mathbf{z} \rangle_A + \langle \mathbf{y}, \mathbf{z} \rangle_A.$$

Linearity in the second argument follows from symmetry.

3. **Positive definiteness:** (3 points) For $\mathbf{x} \in \mathbb{R}^3$,

$$\langle \mathbf{x}, \mathbf{x} \rangle_A = \mathbf{x}^\top A \mathbf{x} = 2x_1^2 + 3x_2^2 + 2x_3^2 + 2x_1x_2 + 2x_2x_3.$$

We show that A is positive definite using **Sylvester's criterion**. The leading principal minors are:

$$\Delta_1 = |2| = 2 > 0, \quad \Delta_2 = \det \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} = 6 - 1 = 5 > 0, \quad \Delta_3 = \det A.$$

Compute $\det A$ by expansion along the first row:

$$\det A = 2 \begin{vmatrix} 3 & 1 \\ 1 & 2 \end{vmatrix} - 1 \begin{vmatrix} 1 & 1 \\ 0 & 2 \end{vmatrix} = 2(6 - 1) - 1(2 - 0) = 10 - 2 = 8 > 0.$$

All leading principal minors are positive, so A is positive definite. Hence $\langle \mathbf{x}, \mathbf{x} \rangle_A > 0$ for all $\mathbf{x} \neq \mathbf{0}$, and $\langle \mathbf{x}, \mathbf{x} \rangle_A = 0$ iff $\mathbf{x} = \mathbf{0}$.

Thus $\langle \cdot, \cdot \rangle_A$ is an inner product on $\mathbb{R}^3$.

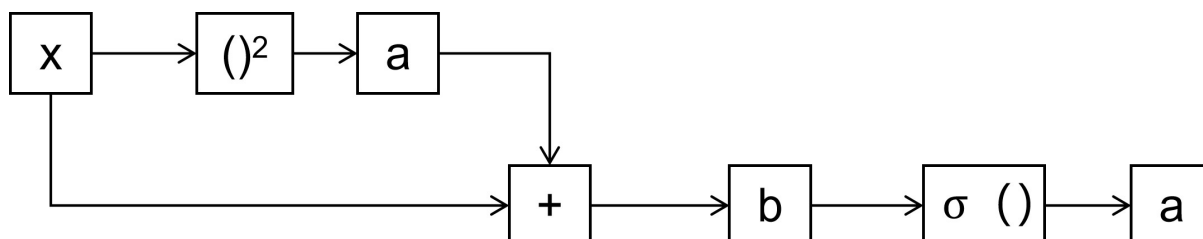
Induced norm: (1 point) Every inner product defines a norm by $\|\mathbf{x}\|_A = \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle_A}$.

The norm axioms (non-negativity, absolute homogeneity, triangle inequality) follow automatically from the inner product properties. Therefore $\|\cdot\|_A$ is a norm on $\mathbb{R}^3$.

Question 5

- (a) 3 points total:

- 1 point: Correctly identify intermediate variables: $a = x^2$, $b = a + x$, $f = \sigma(b)$
- 2 points: Draw a clear directed acyclic graph (DAG) showing the dependencies between variables



- (b) 3 points total:

- 1 point: $a = x^2 = 2^2 = 4$
- 1 point: $b = a + x = 4 + 2 = 6$
- 1 points: $f = \sigma(6) = \frac{1}{1+e^{-6}} \approx 0.9975$

- (c) 4 points total:

- 2 point: Recall the derivative of sigmoid: $\frac{\partial f}{\partial b} = \sigma(b)(1 - \sigma(b))$, $\frac{\partial b}{\partial a} = 1$, $\frac{\partial a}{\partial x} = 2x$, and $\frac{\partial b}{\partial x} = 1$.
- 2 point:

$$\begin{aligned}
 \frac{\partial f}{\partial x} &= \frac{\partial f}{\partial b} \cdot \frac{\partial b}{\partial x} + \frac{\partial f}{\partial b} \cdot \frac{\partial b}{\partial a} \cdot \frac{\partial a}{\partial x} \\
 &= \sigma(b)(1 - \sigma(b)) \cdot 1 + \sigma(b)(1 - \sigma(b)) \cdot 1 \cdot 2x \\
 &= 0.9975 \cdot (1 - 0.9975) + 0.9975 \cdot (1 - 0.9975) \cdot 4 = 0.0125
 \end{aligned}$$

Question 6

Solution: (a) The conditions defining U form a homogeneous linear system. Write the coefficient matrix and row-reduce:

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_2 - R_1} \begin{bmatrix} 1 & 1 & 1 & 1 \\ y0 & -2 & 0 & -2 \end{bmatrix} \xrightarrow{R_2 \leftarrow -\frac{1}{2}R_2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix}.$$

Then eliminate above the pivot in column 2: $R_1 \leftarrow R_1 - R_2$:

$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}.$$

This is the reduced row-echelon form (2 points). Pivot columns are 1 and 2; free variables are x_3 and x_4 .

Let $x_3 = s$, $x_4 = t$ ($s, t \in \mathbb{R}$). The equations become

$$x_1 + s = 0 \Rightarrow x_1 = -s, \quad x_2 + t = 0 \Rightarrow x_2 = -t.$$

(2 points) Thus every $\mathbf{x} \in U$ is

$$\mathbf{x} = \begin{bmatrix} -s \\ -t \\ s \\ t \end{bmatrix} = s \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ -1 \\ 0 \\ 1 \end{bmatrix}.$$

(1 point) The two vectors

$$\mathbf{v}_1 = \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ -1 \\ 0 \\ 1 \end{bmatrix}$$

span U and are linearly independent (they are not multiples of each other). Hence $\{\mathbf{v}_1, \mathbf{v}_2\}$ is a basis for U , and $\dim(U) = 2$. (1 point)

(b) Let the basis vectors be $\mathbf{v}_1$ and $\mathbf{v}_2$. We apply the Gram-Schmidt process to find orthogonal vectors $\mathbf{u}_1, \mathbf{u}_2$.

Step 1: Set the first orthogonal vector.

$$\mathbf{u}_1 = \mathbf{v}_1 = \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

(1 point)

Step 2: Find the second orthogonal vector.

$$\mathbf{u}_2 = \mathbf{v}_2 - \frac{\mathbf{v}_2 \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1} \mathbf{u}_1$$

Notice that $\mathbf{v}_2 \cdot \mathbf{u}_1 = (0)(-1) + (-1)(0) + (0)(1) + (1)(0) = 0$. Since the dot product is zero, the original vectors are already orthogonal.

$$\mathbf{u}_2 = \mathbf{v}_2 = \begin{bmatrix} 0 \\ -1 \\ 0 \\ 1 \end{bmatrix}$$

(2 points)

Step 3: Normalize to find the orthonormal basis $\{\mathbf{w}_1, \mathbf{w}_2\}$.

$$\|\mathbf{u}_1\| = \sqrt{(-1)^2 + 1^2} = \sqrt{2}, \quad \|\mathbf{u}_2\| = \sqrt{(-1)^2 + 1^2} = \sqrt{2}$$

$$\mathbf{w}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{w}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ -1 \\ 0 \\ 1 \end{bmatrix}$$

(1 point)

The orthonormal basis for U is $\{\mathbf{w}_1, \mathbf{w}_2\}$.